

FT85RS With Water Cooling Enclosure Manual V1.5

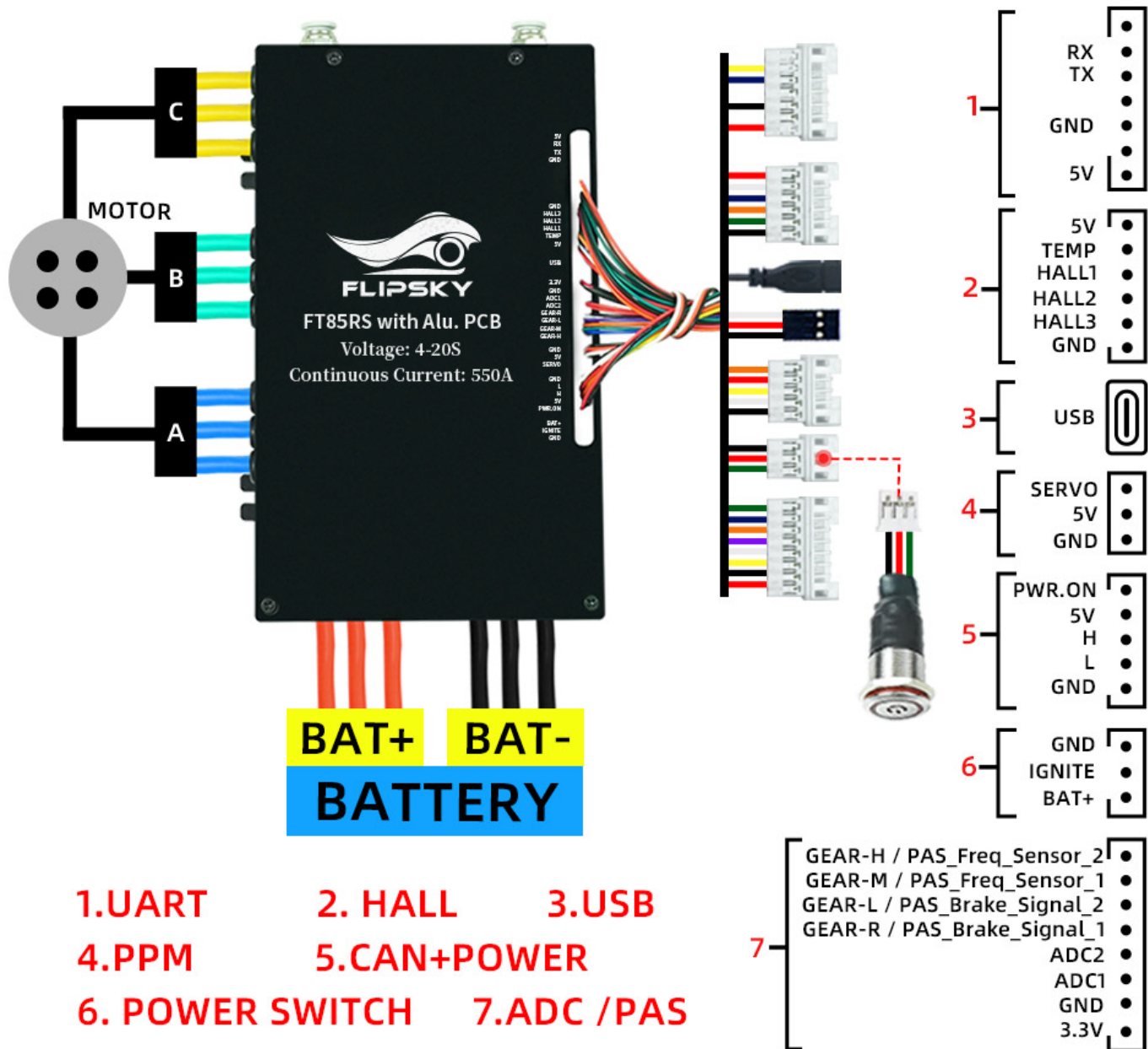
Specifications :

- Firmware: V1.5 or above (firmware update via Flipsky ESC Tool supported)
- Voltage: 14V-85V (safe for 4S-20S)
- Continuous current: 50V/550A; 75V/450A (Values depend on the mounting, ambient temperature and air/water circulation around the device!);
- Peak Current: 1000A
- BEC:5V@1A
- Motor/power Wire Size:8 AWG (3pcs combine)
- Programmable Tool: Flipsky ESC TOOL
- PWM Frequency Control: YES
- Multiple Speed Gear Limit Control:YES (UART / ADC)
- Regenerative Capacity: YES
- Communication Ports: USB, CAN, UART, PPM, ADC
- Input Signals Support: PPM, ADC, PAS, UART(compatible with Flipsky remote such as VX1, VX1Pro, VX2, VX3, VX3Pro, VX4Pro)
- Startup Modes Support: IPDS (sensorless), HFPI (sensorless), HALL (sensored), ENCODER(ABI Encoder, AS5047 Absolute Type Encoder SPI Port), OPEN (Sensorless Open-Loop Startup)
- Traction Control: YES
- Multiple Speed Gear levels: Low, Medium, High, Reverse
- Switch Support: Power on/off button switch or key switch
- Size: L227.5*W140*H54mm(with water cooling enclosure)

Features:

- Adopting ARM Cortex-M4 structure with the max 240MHz , improves PWM switching frequency and the motor RPM.
- Programmable via Flipsky ESC Tool to be compatible with multi-types of motor, different battery voltages and many control ways.
- Programmable via Flipsky remote controller VX4Pro (remote firmware version: VX4Pro_V1.1 or above).
- Supporting motor FOC with low working noise and smooth starting.
- Supporting Real-time data and fault alerts.
- Supporting adaptive variable PWM frequency control.
- Supporting sensor mode(HALL, ENCODER) and sensorless mode(IPDS, HFPI, OPEN).
- Supporting control types: Current, Duty Cycle, Speed control, Position Control
- Protection functions: low and high voltage protection, over-current protection, temperature abnormally protection, MOSFET over-temperature and over current protection, battery over discharge protecting, motor over-temperature protection.
- Supporting multiple typed of motor temperature sensors such as NTC-10K-25℃ , <https://flipsky.net/> NTC-10K-100℃ , PT1000, PTC-1K-100℃ , KTY and so on. E-mail:flipskytech@outlook.com

Wiring diagram:



To download the Flipsky ESC
Tool and User Manual,
please scan the QR code:



FT85RS铝基水冷版规格书（版本V1.5）

规格参数:

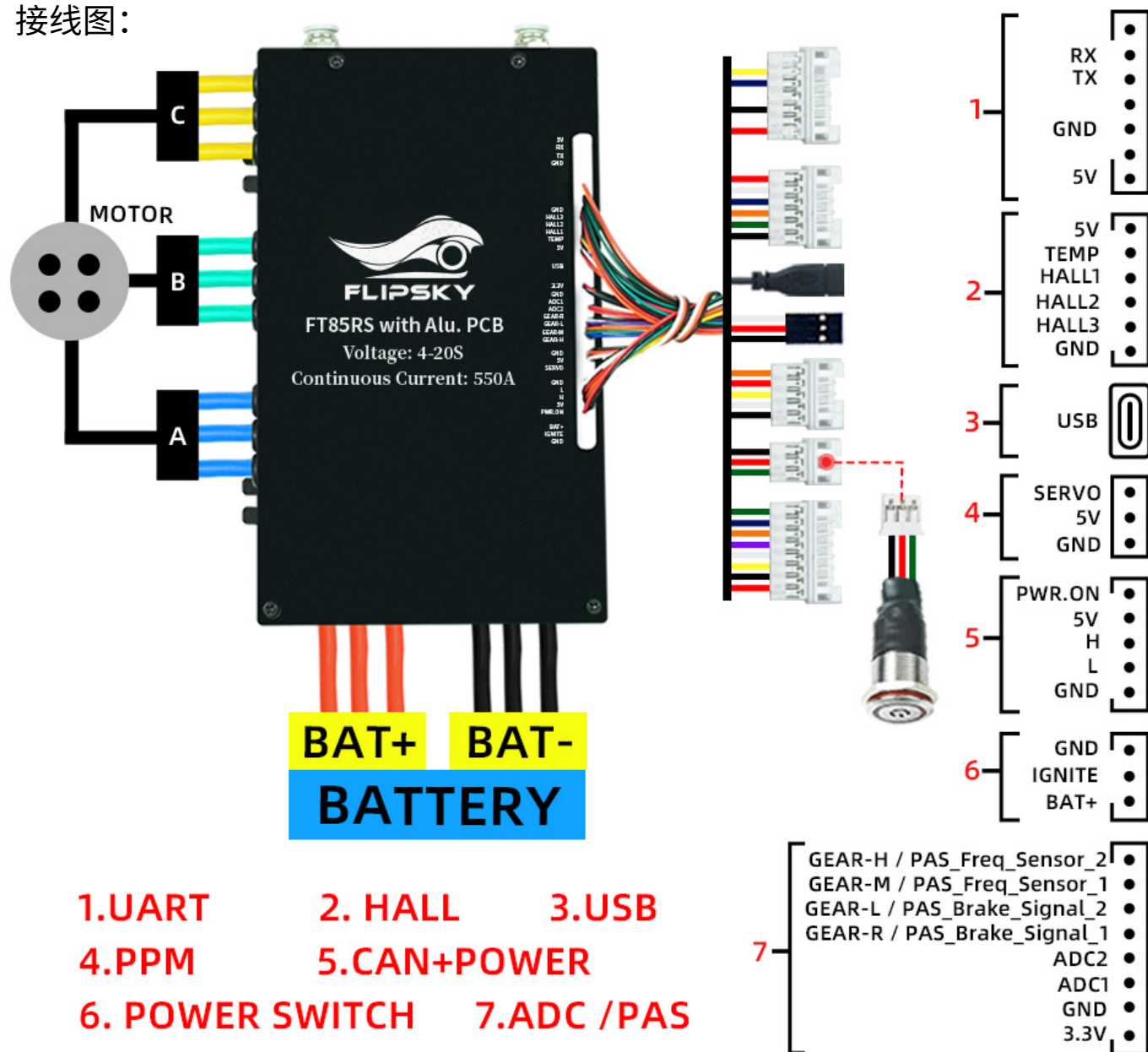
- 固件：V1.5及以上，支持通过Flipsky ESC Tool升级
- 电压：14V-85V (安全范围：4S-20S)
- 持续电流：50V/550A 75V/450A （持续时间取决于外部散热情况）
- 峰值电流：1000A
- BEC:5V@1A
- 调参工具：Flipsky ESC TOOL
- PWM变频控制：YES
- 多挡位限速控制：YES(UART、ADC)
- 再生制动：YES
- 通讯端口：USB, CAN, UART, PPM, ADC
- 控制信号类型：PPM, ADC, PAS, UART （目前兼容 VX1、VX1Pro、VX2、VX3、VX3Pro、VX4Pro 等遥控器）
- 启动模式：IPDS(无感轻载), HFPI(无感重载), HALL, ENCODER（ABI编码器，AS5047绝对式编码器SPI接口），OPEN(无感开环启动)
- 牵引力控制：YES
- 速度档位：低速档，中速档，高速档，倒车档
- 集成了自锁开关或电门锁开关
- 电源线缆：8 AWG（3组）
- 电机线缆：8 AWG（3组）
- 尺寸：长227.5 mm、宽140 mm、高54 mm（带水冷外壳）

产品特点:

- 采用ARM Cortex-M4架构，高达240MHz主频，可提高PWM控制频率和电机转速上限
- 可通过Flipsky ESC Tool编程，兼容多种电机、不同电池电压和多种控制方式
- 支持 Flipsky VX4Pro 遥控器调参（固件版本：VX4Pro_V1.1 以上版本）；
- 支持 FOC，工作噪音低，启动平稳；
- 支持实时数据和故障报警支持自适应PWM变频控制
- 支持传感器模式（HALL、ENCODER）和无传感器模式（IPDS、HFPI、OPEN）
- 支持控制类型：电流（Current），占空比（Duty Cycle），速度控制（Speed Control），位置控制（Position Control）
- 保护功能：低压保护；高压保护；电机过温保护；MOS管过流保护；电池过放保护；MOS管过温保护；

- 支持多种类型电机温度传感器类型，包括NTC-10K-25°C、NTC-10K-100°C、PT1000、PTC-1K-100°C、KTY系列等多种类型温度传感器

接线图：



Flipsky ESC TOOL
软件+中文用户手册
下载二维码：

